

Multi-Factor Driver Authentication System

Complete Technical Report

Phases 1 – 5

Face Detection · Fingerprint · Relay & Ignition · GSM/SMS · Integration



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Executive Summary

This report consolidates the full technical documentation of the Multi-Factor Driver Authentication System — a Raspberry Pi-based unit that secures a vehicle's ignition behind two biometric factors and is fully operable by SMS. The work is organised into five phases, each delivering an independently testable subsystem:

- Phase 1 — Face Detection: Haar detection, multi-pose registration, LBPH recognition, and auto-lighting.
- Phase 2 — Fingerprint Authentication: R307S enrollment, verification, and the dual-factor decision.
- Phase 3 — Relay & Ignition Control: 3-channel relay, timer-latch power hold, and ignition gating.
- Phase 4 — GSM/SMS Remote Control: SIM800 command/alert layer over the cellular network.
- Phase 5 — System Integration & Deployment: the Main.py state machine, kiosk UI, and lifecycle.

Each part below corresponds to one phase. Phase 1 includes the captured screenshots; remaining placeholders are provided for additional photos.

Phase 1

Face Detection

1. Overview

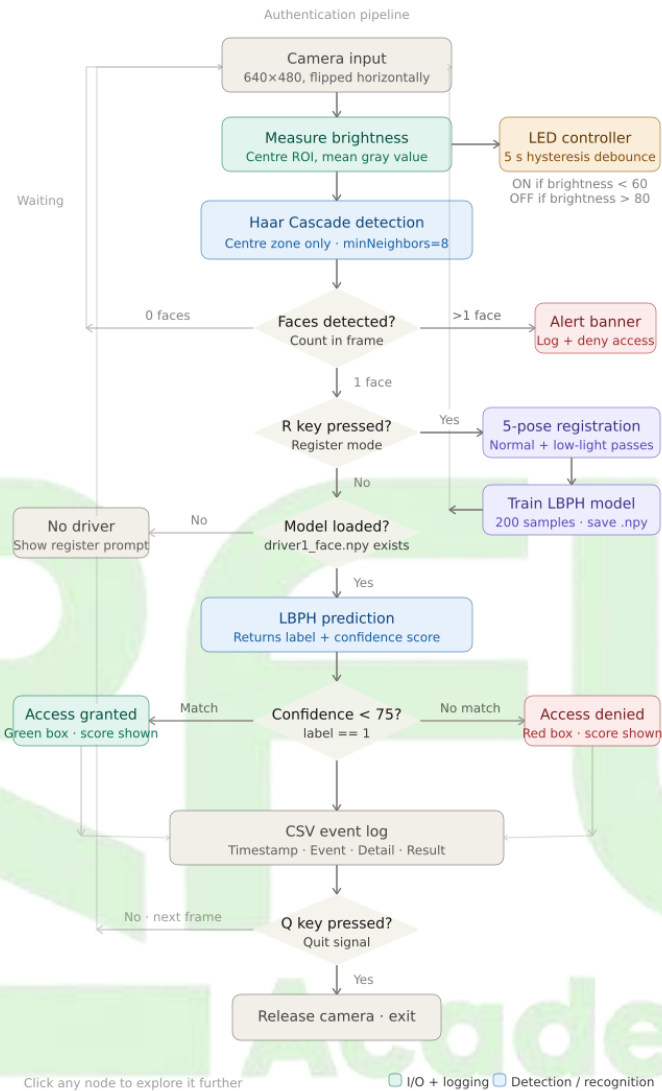
Phase 1 implements the core computer-vision pipeline: face detection, driver registration, real-time face recognition, and automatic LED lighting control. It is software-only and runs on a standard PC or Raspberry Pi with an attached camera.

1.1 Key Features

- Strict Haar Cascade face detection (centre-zone only, high confidence)
- iPhone-style multi-pose registration — 5 poses × 2 lighting × 20 samples = 200 samples
- LBPH face recognizer; multiple-face alert; debounced LED control; CSV logging

2. System Architecture

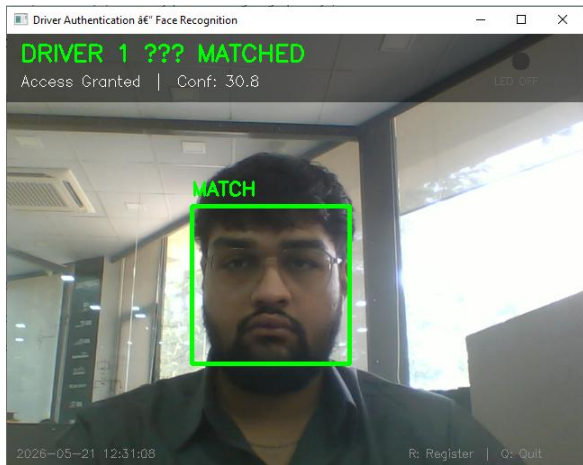
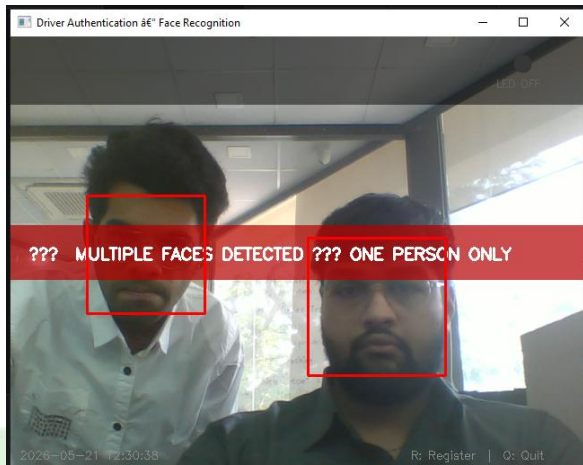
Module	Responsibility
Face Detection	Centre-zone Haar detection (minNeighbors=8, minSize=100×100).
Registration	5 poses × 2 lighting × 20 frames; trains LBPH.
Authentication	Compares live ROI to the LBPH model against a confidence threshold.
LED Controller	Central-ROI brightness with a debounced relay.



System flow diagram — the full authentication pipeline

3. Face Detection

Constant	Value	Description
DETECT_SCALE	1.1	Image-pyramid scale factor
DETECT_MIN_NEIGHBORS	8	Higher = fewer false positives
DETECT_MIN_SIZE	(100, 100)	Minimum face size in pixels
DETECT_ZONE_X / Y	80 / 60 px	Detection-zone edge insets

Constant	Value	Description
		
	Single face — normal detection	Multiple-face alert — access blocked

During registration, detection is relaxed ($\text{minNeighbors}=4$, $\text{minSize}=60 \times 60$) to allow angled poses.

4. Driver Registration

Sample Count

5 poses $\times$ 2 lighting $\times$ 20 samples = 200 training samples stored in driver1_face.npy.

1

Look straight at the camera

Baseline frontal view — normal + low light

2

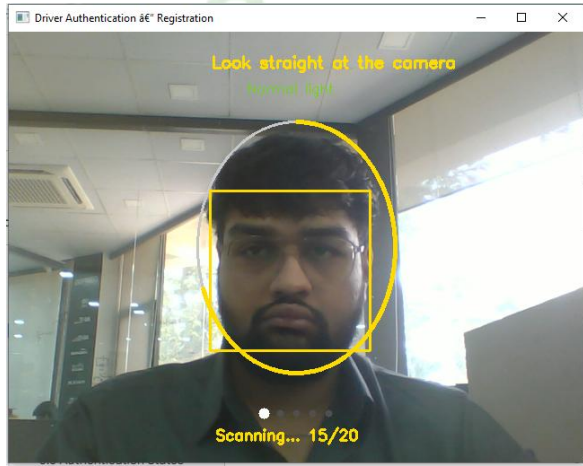
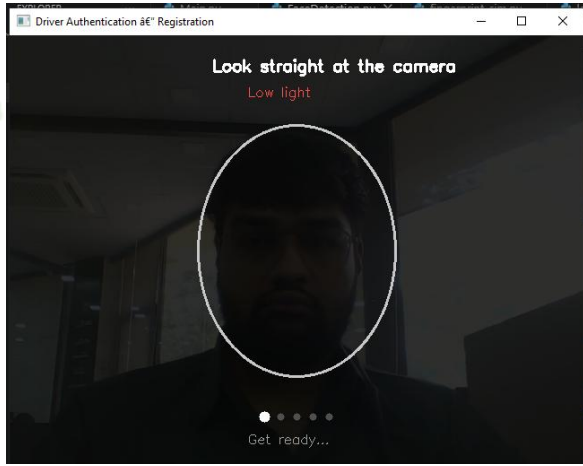
Turn head RIGHT / LEFT

Profile variations

3

Tilt head UP / DOWN

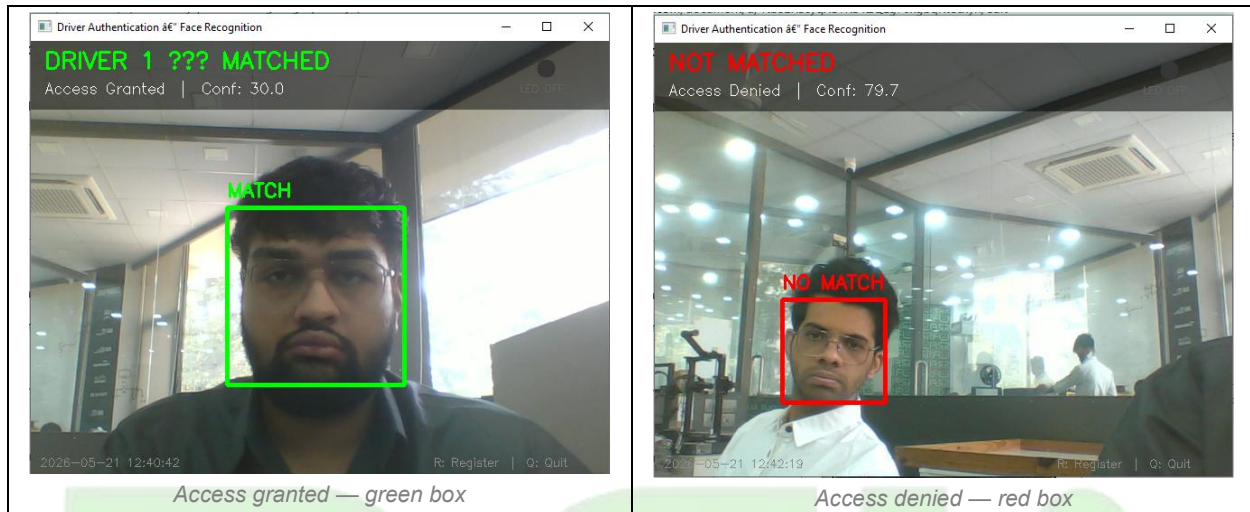
Gaze variations

	
Step 1 — normal light	Step 1 — low light



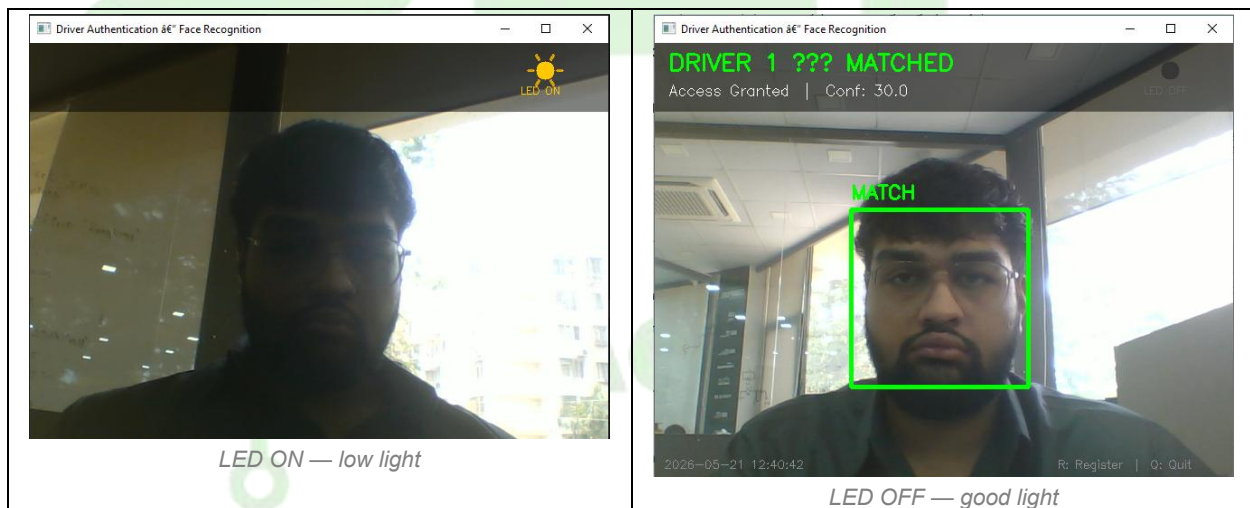
5. Authentication

LBPH returns a confidence value; lower is a better match. `MATCH_THRESHOLD = 75`: granted if `label == 1` and `confidence < 75`.



6. LED Auto-Lighting & Logging

Brightness is measured in a central ROI; the LED relay debounces with a 5-second confirmation (ON < 60, OFF > 80). All events are logged to auth_log.csv.




```
auth_log.csv > data
1   Timestamp,Event,Detail,Result
9083 2026-05-21 12:45:52,FACE_MATCH,conf=43.6,GRANTED
9084 2026-05-21 12:45:52,FACE_MATCH,conf=43.8,GRANTED
9085 2026-05-21 12:45:52,FACE_MATCH,conf=43.8,GRANTED
9086 2026-05-21 12:45:52,FACE_MATCH,conf=44.5,GRANTED
9087 2026-05-21 12:45:52,FACE_MATCH,conf=45.2,GRANTED
9088 2026-05-21 12:45:52,FACE_MATCH,conf=43.4,GRANTED
9089 2026-05-21 12:45:52,FACE_MATCH,conf=45.0,GRANTED
9090 2026-05-21 12:45:52,FACE_MATCH,conf=45.0,GRANTED
9091 2026-05-21 12:45:52,FACE_MATCH,conf=44.5,GRANTED
9092 2026-05-21 12:45:52,FACE_MATCH,conf=46.1,GRANTED
9093 2026-05-21 12:45:53,FACE_MATCH,conf=44.8,GRANTED
9094 2026-05-21 12:45:53,FACE_MATCH,conf=47.9,GRANTED
9095 2026-05-21 12:45:53,FACE_MATCH,conf=47.8,GRANTED
9096 2026-05-21 12:45:53,FACE_MATCH,conf=47.3,GRANTED
9097 2026-05-21 12:45:53,FACE_MATCH,conf=49.3,GRANTED
9098 2026-05-21 12:45:53,FACE_MATCH,conf=47.1,GRANTED
9099 2026-05-21 12:45:53,FACE_MATCH,conf=47.9,GRANTED
9100 2026-05-21 12:45:53,FACE_MATCH,conf=48.7,GRANTED
9101 2026-05-21 12:45:53,FACE_MATCH,conf=51.2,GRANTED
9102 2026-05-21 12:45:54,FACE_MATCH,conf=50.1,GRANTED
9103 2026-05-21 12:45:54,FACE_MATCH,conf=54.2,GRANTED
9104 2026-05-21 12:46:12,FACE_NO_MATCH,conf=77.0,DENIED
9105 2026-05-21 12:46:12,FACE_MATCH,conf=69.9,GRANTED
9106 2026-05-21 12:46:12,FACE_MATCH,conf=71.7,GRANTED
9107 2026-05-21 12:46:12,FACE_MATCH,conf=67.9,GRANTED
9108 2026-05-21 12:46:13,FACE_MATCH,conf=67.9,GRANTED
9109 2026-05-21 12:46:13,FACE_MATCH,conf=67.6,GRANTED
9110 2026-05-21 12:46:13,FACE_MATCH,conf=66.3,GRANTED
9111 2026-05-21 12:46:13,FACE_MATCH,conf=65.7,GRANTED
9112 2026-05-21 12:46:17,FACE_MATCH,conf=65.1,GRANTED
9113 2026-05-21 12:46:18,FACE_MATCH,conf=71.9,GRANTED
9114 2026-05-21 12:47:57,FACE_NO_MATCH,conf=76.5,DENIED
```

auth_log.csv — timestamped event log

Phase 2

Fingerprint Authentication

1. Overview

Phase 2 adds fingerprint authentication as a second factor. Face recognition is combined with a live R307S optical fingerprint scan, so the vehicle is only authorised when both factors agree.

1.1 Key Features

- R307S over UART (/dev/ttyAMA0, 57600 baud) via pyfingerprint
- Two-scan guided enrollment; on-sensor template at slot #1 with a disk flag
- Dual-factor decision: GRANTED only when face AND fingerprint match
- Scan states WAITING / SCANNING / LIFT / DONE; simulation mode for development

2. Hardware & Wiring

Wire	Signal	Pi Pin
Red / Black	VCC / GND	Pin 4 (5V) / Pin 6 (GND)
Yellow / Green	TXD / RXD	Pin 10 (RX) / Pin 8 (TX)
Blue	TOUCH	Pin 11 (GPIO17, active-low)
White	WAKEUP	Leave floating

Serial Setup

raspi-config → Serial Port: login shell = No, hardware = Yes, reboot. Sensor port /dev/ttyAMA0.

3. Enrollment & Verification

Enroll Once, Verify Forever

The template lives at sensor slot #1 and fingerprint_enrolled.flag is written to disk, surviving restarts.

1

Place finger on the sensor

First scan into CharBuffer 1

2

Place the SAME finger again

Second scan confirms the first

3

Template created and stored

Stored at slot #1; flag saved

```
result = f.searchTemplate(); slot, score = result[0], result[1]
if slot == -1: return False, conf          # no match
confidence = round(min(99.9, score/2.0),1) # MATCH
```

4. Dual-Factor Decision

```
final = "GRANTED" if (face_matched and fp_matched) else "DENIED"
```

Face-Only Auto-Check

A passive auto-check decides on the face alone for presence detection — never to grant ignition. Full authentication always requires both factors.



Phase 3

Relay & Ignition Control

1. Overview

Phase 3 connects the authentication logic to hardware through a 3-channel relay: a timer-latch relay that keeps the Pi powered, the LED relay, and the ignition relay enabled only on a dual-factor GRANTED decision.

2. Relay Channels

Channel	BCM Pin	Polarity	Purpose
Timer Latch	GPIO23	Active-HIGH	Self-holding power latch energised on boot
LED Light	GPIO27	Active-LOW	Cabin / sensor lighting (auto-brightness)
Ignition	GPIO22	Active-HIGH	Ignition enable, gated on GRANTED

Per-Pin Polarity

ACTIVE_LOW_PINS = {PIN_LED_RELAY}; only the LED channel energises on LOW. `_set_pin()` inverts the level automatically.

3. Timer-Latch Power Hold

The Pi energises GPIO23 on boot; the latch then holds its own supply on, keeping the Pi powered past the external boot-timer window.

```
def relay_init(): _init_gpio(); timer_latch_on() # GPIO23 HIGH
def relay_shutdown(): ignition_off(); led_relay_off()
# latch left ON; never GPIO.cleanup() (would cut power)
```

Critical — Never Call `GPIO.cleanup()`

`cleanup()` floats GPIO23, de-energising the latch and instantly cutting power. Shutdown releases only ignition + LED.

4. Ignition & LED

Trigger	Action	GPIO22
Full auth → GRANTED	<code>ignition_on()</code>	HIGH
DENIED / auto-check DENIED	<code>ignition_off()</code>	LOW
TURN OFF (remote)	<code>ignition_off()</code>	LOW

The LED auto-lighting now drives the active-low GPIO27 relay, with an SMS LED ON / LED OFF override.

Phase 4

GSM / SMS Remote Control

1. Overview

Phase 4 makes the system remotely operable over the cellular network. A SIM800 GSM module lets the owner control and monitor the vehicle entirely by SMS, and pushes alerts back on every access decision.

Active Hardware Status

The SIM800 is the main open hardware issue: it may not respond to AT commands until a 2.8 V (module) vs 3.3 V (Pi) UART logic-level concern is resolved. See Troubleshooting.

2. Hardware & Architecture

Setting	Value	Notes
Serial port	/dev/ttyAMA2	UART2 overlay (separate from R307S)
Baud / Mode	115200 / text	AT+CMGF=1 SMS text mode
Country code	+91	Prepended to the 10-digit number

A persistent GsmPoller thread holds the port open and polls every 1.5 s; outbound sends use separate short-lived connections so they never block polling.

Why Messages Are Deleted Immediately

Each SMS from AT+CMGL is deleted with AT+CMGD as soon as it is read — SIM slot indices get reused, so a 'seen' set would skip reused slots and fill storage.

3. Command Reference

Command	Effect
AUTH	Run full dual-factor authentication
REGISTER FACE / FP / PHONE	Enroll face / fingerprint / owner number
STATUS / LOG / HELP	Status summary / last 5 logs / command list
TURN OFF	Kill the ignition relay
LED ON / OFF, WAKE UP	Lighting override; wake from sleep
RESET / RESET CONFIRM <pwd>	Begin / confirm a password-protected data reset

4. Troubleshooting (hardware first)

- Logic levels — SIM800 UART $\approx$ 2.8 V vs Pi 3.3 V; a level shifter on TX/RX may be needed.
- Power — hold under $\sim$ 2 A transmit bursts; brownouts drop the module mid-command.
- Signal — AT+CPIN? READY, AT+CSQ not 99, AT+CREG? 0,1 or 0,5. Test/GSM_test.py runs the full sequence.

Phase 5

System Integration & Deployment

1. Overview

Phase 5 ties every subsystem from Phases 1–4 into a single always-on application driven by a state machine and presented as a full-screen kiosk. Main.py is the orchestrator: camera, UI, state, and SMS routing.

2. Architecture & Threading

Module	Role	Phase
FaceDetection / fingerprint_sim	Recognition + fingerprint	1 / 2
relay_controller / sms_alert	Relays + GSM SMS	3 / 4
logger / first_boot	Logs + setup wizard	1–5 / 5
Main.py	State machine, UI, lifecycle, dispatch	5

The main thread runs the render loop, state machine, and SMS handling (all shared-state mutation here); workers and the GsmPoller run in the background.

3. State Machine

State	Meaning
IDLE / SLEEPING	Live preview / dimmed after inactivity (SMS still active)
FIRST_BOOT	Guided setup wizard over SMS
REG_FACE / REG_FP	Registration / enrollment in progress
AUTHING / RESULT	Authentication running / showing the outcome
RESETTING	Data reset being executed

4. First-Boot, Kiosk & Lifecycle

Setup Step	Required?	Completed By
password / face / fingerprint	Required	SET PASSWORD / REGISTER FACE / REGISTER FP
phone	Optional	REGISTER PHONE <num>

The UI is one full-screen OpenCV window, display-only, auto-detecting the GTK/Wayland/X11 backend. On boot relay_init() energises the timer latch; on exit relay_shutdown() releases ignition + LED but leaves the latch ON.

Power-Safe Lifecycle

The timer latch is energised first on boot and left on through shutdown, guaranteeing stable power for the entire run.

```
pip install opencv-contrib-python numpy picamera2 \
           pyfingerprint pyserial RPi.GPIO --break-system-packages
python3 Main.py
```

The in-app kiosk is implemented; OS-level autostart at session start (Wayland/labwc) is the final commissioning step.

With Phase 5 the system is functionally complete: a single Raspberry Pi unit that recognises a driver's face and fingerprint, gates ignition through a relay, holds its own power, and is fully operable by SMS — an unattended full-screen kiosk documented end to end across the five phases.

